

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |
Kent, OH - Open to relocation to Paris (visa sponsorship required)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) delivering enterprise AI solutions end-to-end in compressed timelines - RAG systems, agentic LLM workflows, and production inference services. Master's degree (Kent State University). Track record of measurable outcomes: ~35% retrieval precision improvement, >30% hallucination reduction, ~30% fewer post-deployment defects. Comfortable with full-stack delivery: Python (FastAPI, Flask), React/TypeScript, Docker, and LLM orchestration (LangChain, LlamaIndex). Strong client communication skills from regulated healthcare environments. Seeking Applied AI Engineer, Prototyping role at Mistral AI - bringing production-grade AI delivery discipline to 4-8 week enterprise prototype cycles.

CORE COMPETENCIES

LLM Application Development • Retrieval-Augmented Generation (RAG) • Agentic Workflows • Prompt Engineering • Structured Outputs • Evaluation Pipelines • Python / FastAPI / Flask • React / TypeScript • LangChain / LlamaIndex • Hugging Face Transformers / PyTorch • Docker / CI-CD • System Design • Enterprise AI Delivery • Technical Stakeholder Communication

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare AI, USA

- Built **production RAG systems** for clinical knowledge retrieval: recursive semantic chunking, transformer-based embeddings, retrieval evaluation loop - improved contextual retrieval precision ~**35%**; retrieval-grounded response alignment exceeded **90%**.
- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care navigation, documentation clarification) using LangChain - tool discipline, grounding rules, structured reasoning; reduced hallucination >**30%** and improved agent stability ~**25%**.
- Shipped **FastAPI/Flask RESTful inference services** containerized with Docker; structured logging, load simulation, and monitoring - reduced post-deployment defects ~**30%**.
- Built preprocessing pipelines (Pandas, NumPy) for EHR extracts; raised dataset reliability above **98%** and cut downstream model instability ~**40%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Hyderabad, India

- Built **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions - reduced manual review time **60-70%**; ~**85% alignment** with human-curated highlights.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval) - reduced hallucinations ~**30%** and raised contextual answer accuracy >**85%**.
- Deployed the stack as a **Flask API with minimal web UI** - non-technical staff used the system independently without training.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built **Jenkins CI/CD pipelines** for SQL schema updates and stored procedure deployments - reduced deployment errors >**30%**, improved release cycles ~**35-40%**.
- Designed **Grafana dashboards** for performance monitoring (SQL Server DMVs, long-running queries, deadlocks) - cut incident recurrence ~**25%**.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

LangChain, LlamaIndex, FastAPI, Docker, Python

Manga Lens - Chrome Extension for Real-Time AI Manga Translation shipped - Chrome Web Store

Full-stack browser extension - Manifest V3, React frontend, service workers. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama local). Multi-provider payload handling, viewport-slice screenshot pipeline, 7-day translation cache, per-domain configs for 29 manga sites.

TypeScript, React, Python, Manifest V3, multi-provider AI APIs

Dream Decoder - AI-Powered Multimodal Creative Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app with coordinated multimodal APIs (Perplexity Sonar + Replicate image models). Introduced **structured prompt transformation layers** - improved contextual alignment ~**30%** and first-pass image success rate ~**25-30%**.

FastAPI, React, TypeScript, Tailwind, multi-modal AI APIs

Agentic Pixel Character Synthesis & Animation Engine

Generative AI system for identity-consistent, pose-controlled pixel character synthesis. **LLM-based agent orchestrator** decomposes high-level prompts into generation tasks. Stable Diffusion fine-tuning + LoRA + ControlNet + OpenPose/MediaPipe for pose conditioning.

PyTorch, Hugging Face Diffusers, FastAPI, Docker, LLM orchestration

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, SQL (PostgreSQL, T-SQL, SQLite)

LLM & GenAI: LangChain, LlamaIndex, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, YOLOv8, OpenCV

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment

Work Authorization: F-1 OPT (US-authorized); requires visa sponsorship for France/EU